

Secondary Protein Databases

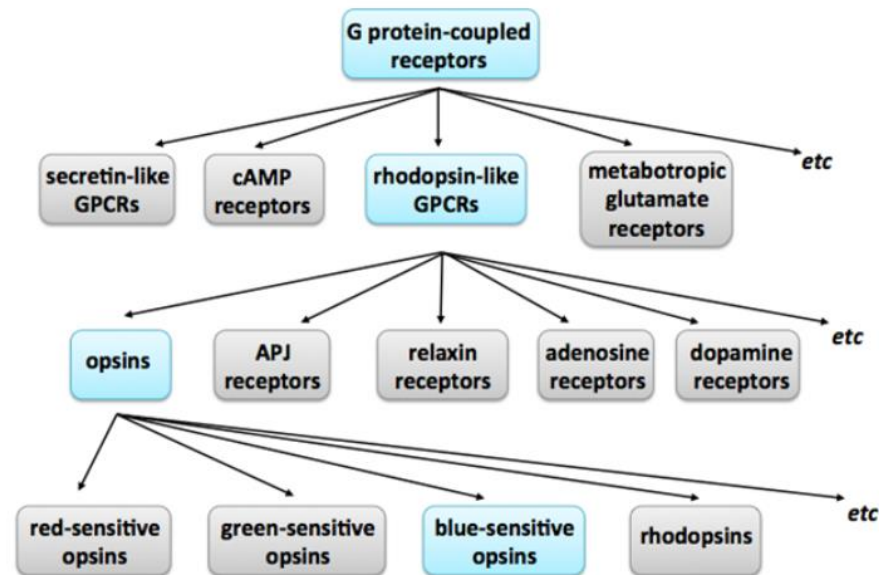
Shatakshi Kulkarni

Introduction

- ❑ Secondary databases comprise data derived from the results of analyzing primary data.
- ❑ They are highly curated, often using a complex combination of computational algorithms and manual analysis and interpretation to derive new knowledge from the public record of science.
- ❑ Secondary protein databases offer derived or interpreted data that has been processed or analyzed.
- ❑ E.g., Prosite, Prodom, CDD, InterPro, Pfam (protein family db of alignment & HMM), PIRSF (superfamily classification system)
- ❑ These databases are essential tools in bioinformatics for understanding protein structure, function, and evolution, and they are often used in conjunction with primary databases like PDB (Protein Data Bank) and UniProt.

Introduction

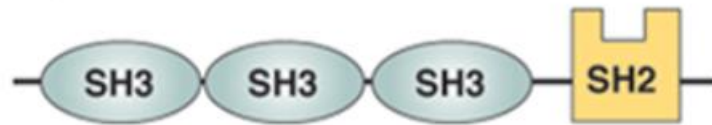
Protein Families: Protein families are often arranged into hierarchies, with proteins that share a common ancestor subdivided into smaller, more closely related groups. The terms superfamily (describing a large group of distantly related proteins) and subfamily (describing a small group of closely related proteins) are sometimes used in this context.



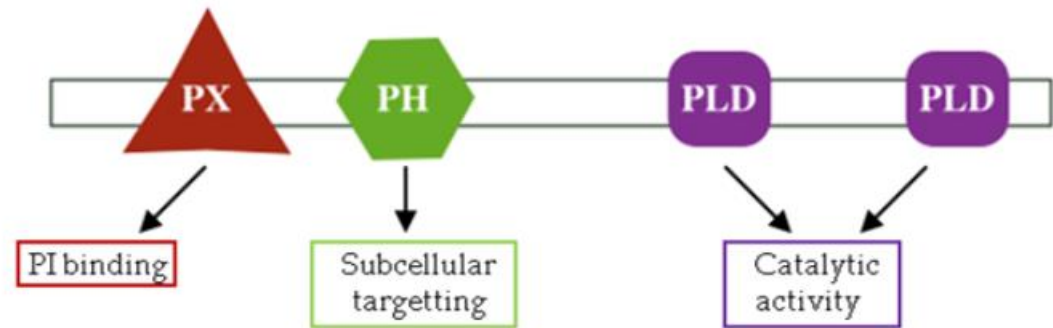
Source (EBI)

Introduction

Protein Domains: Domains are distinct functional and/or structural units in a protein. Usually they are responsible for a particular function or interaction, contributing to the overall role of a protein. Domains may exist in a variety of biological contexts, where similar domains can be found in proteins with different functions.



Multiple domain in NCK protein



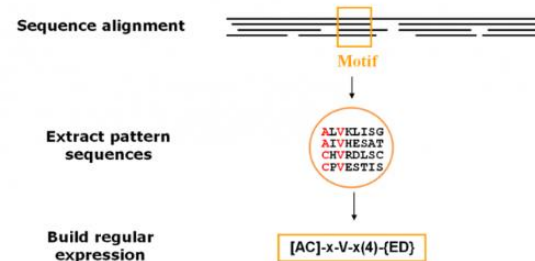
Different domains in one protein

Source (EBI)

Introduction

Patterns:

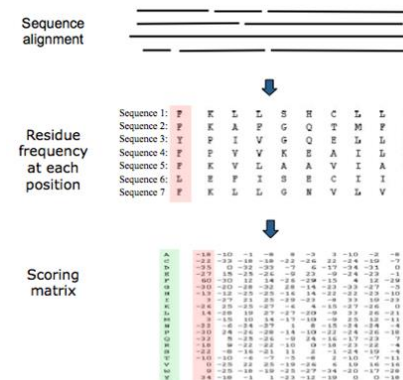
Many important sequence features, such as binding sites or the active sites of enzymes, consist of only a few amino acids that are essential for protein function. Patterns are very good at recognizing such features. They are built by identifying these regions in multiple sequence alignments. The pattern of conservation within the sequence feature is then modelled as a regular expression.



Profile:

Profiles are used to model protein families and domains. They are built by converting multiple sequence alignments into position-specific scoring systems (PSSMs). Amino acids at each position in the alignment are scored according to the frequency with which they occur. Each of the 20 amino acids commonly found in proteins is given a score for each position in the sequence according to the frequency with which they occur in the original alignment.

[Signature types | Protein classification \(ebi.ac.uk\)](http://ebi.ac.uk) : Reference



PROSITE

- ❑ PROSITE is a database of protein families and domains.
- ❑ It consists of entries describing the protein families, domains and functional sites as well as amino acid patterns and profiles in them.
- ❑ Its patterns and profiles can help to determine what is the function of uncharacterized proteins translated from genomic or cDNA sequences.
- ❑ With appropriate computational tools patterns and/or profiles can rapidly and reliably identify which known family of protein (if any) the new sequence belongs to and/or which domain(s) and/or sites it contains.
- ❑ These are manually curated by a team of the Swiss Institute of Bioinformatics and tightly integrated into Swiss-Prot protein annotation.
- ❑ It uses patterns and profiles to predict the function of uncharacterized proteins based on their sequences. These patterns, also known as motifs, arise from specific requirements in protein structure, such as binding properties or enzymatic activity, and are highly conserved across evolution.
- ❑ When a new protein sequence is too distantly related to known proteins to be identified by sequence alignment alone, these motifs can help determine its function by matching it to known protein families or domains.
- ❑ PROSITE was developed to aid in the functional annotation of proteins by identifying these conserved patterns.
- ❑ While some proteins are too diverse to be detected by patterns alone, profiles—techniques based on weight matrices—allow for the detection of more distantly related proteins and domains. Since 1994, PROSITE has focused on expanding its database with profiles to improve its detection capabilities, with contributions from the Swiss Institute of Bioinformatics.

PROSITE

- ❑ PROSITE's uses include identifying possible functions of newly discovered proteins and analysis of known proteins.
- ❑ PROSITE offers tools for protein sequence analysis and motif detection. It is part of the ExPASy proteomics analysis servers.
- ❑ Input: FASTA, UniProtKB accession ID, PDB ID
- ❑ Hits by pattern and profile are provided in the flatfile. Each domain can be studied in detail
- ❑ **Manual:** https://prosite.expasy.org/scanprosite/scanprosite_doc.html#of_miniprofiles

Hits (Similar sequences)

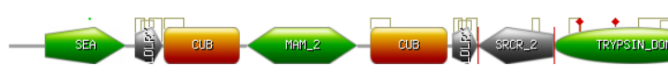
No. of AAs

hits by profiles: [8 hits (by 6 distinct profiles) on 1 sequence]

Upper case represents match positions, lower case insert positions, and the '.' symbol represents deletions relative to the

ruler: 1 100 200 300 400 500 600 700 800 900

P98073
(ENTK_HUMAN)



RecName: Full=Enteropeptidase; EC=3.4.21.9; AltName: Full=Enterokinase; AltName: Full=Serine protease 7; AltName: Full=Enteropeptidase non-catalytic heavy chain; Contains: RecName: Full=Enteropeptidase catalytic light chain

PS50024 SEA SEA domain profile:

54 - 169: score = 32.979

QSHEARATFKITsGVTYNPNLQDKLSVDFKVLAFDLQMQHIDEIFLSSNLKNEYKNSRVLQ
FEN--LIVVFDLFFAQWVSD--ENVKEELIQGLEANKssQLVTFHIDLNSVDILDKLT

Predicted features:

Feature	Start	End	Description	Condition
DOMAIN	54	169	/note="SEA "	[condition: none]
SITE	117	118	/note="Cleavage; by autolysis"	[condition: none]

PS50068 LDLRA_2 LDL-receptor class A (LDLRA) domain profile: Hits of PS50068 on PDB 3D structure x-referen

Domain Sequence

PROSITE

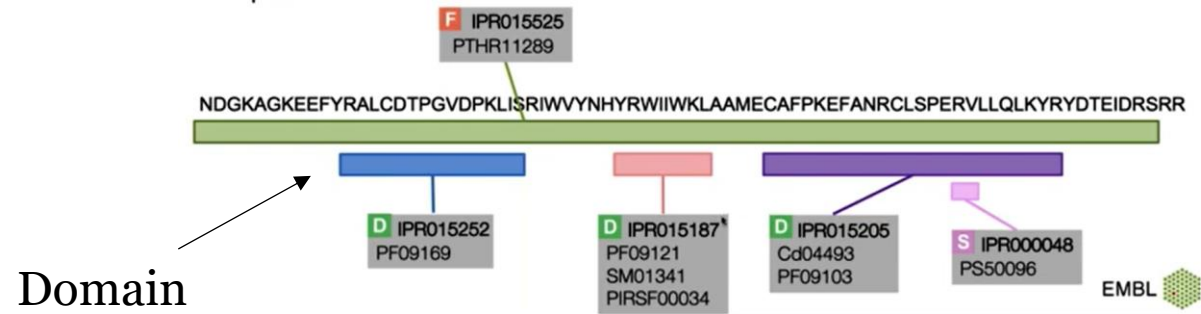
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PRODOR

☐ When Prodom was maintained:

- The ProDom database contains protein domain families generated from the SWISS-PROT database by automated sequence comparisons.
 - It can be searched on the World Wide Web or by E-mail to study domain arrangements within known families or new proteins.
 - Strong emphasis has been put on the graphical user interface which allows for interactive analysis of protein homology relationships.
 - Recent improvements to the server include: ProDom search by keyword; links to PROSITE and PDB entries; more sensitive ProDom similarity search with BLAST or WU-BLAST; alignments of query sequences with homologous ProDom domain families; and links to the SWISS-MODEL server for homology based 3-D domain modelling where possible.
- ☐ It is no longer actively maintained. However, if you still want to explore its contents, you can access the archived version of the ProDom database or use tools like Pfam and InterPro, which have incorporated many of the functions that ProDom once served.

InterPro



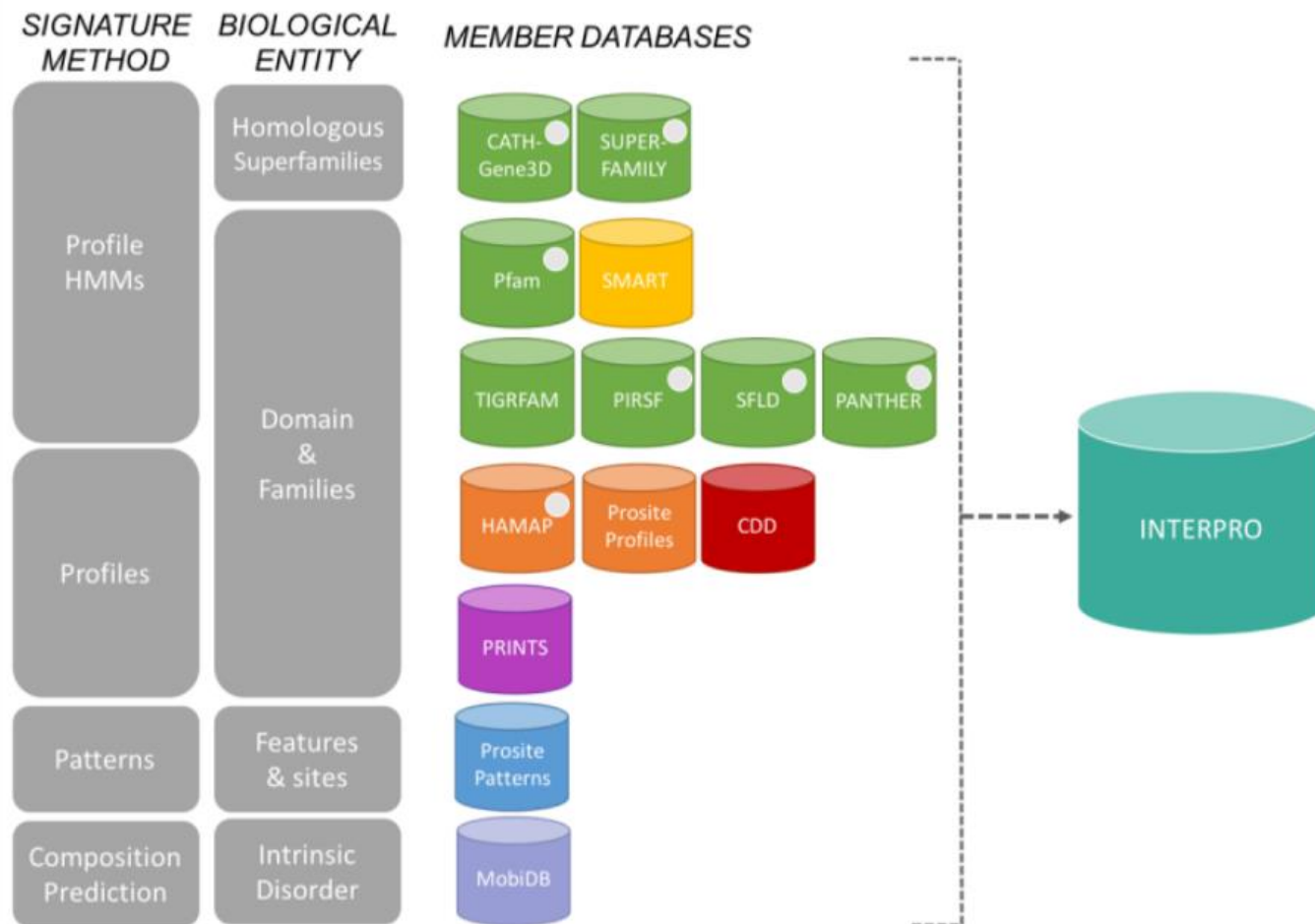
- ❑ InterPro is the main resource for protein classification and functional analysis at the EBI.
- ❑ In InterPro, patterns, profiles, fingerprints and HMMs from a number of different databases are brought together into a single searchable resource, offering convenient access to their predictive capabilities without the need to visit the member databases individually.
- ❑ By combining the different databases and signature types, InterPro capitalizes on their individual strengths, producing a powerful tool for the prediction of protein function.
- ❑ InterPro aims to simplify and rationalize protein sequence analysis for the user by combining and organizing information in a consistent manner, removing redundancy, and adding extensive annotation and useful links about the signatures and the proteins they match.
- ❑ Predicts protein families & presence of domains, important sites

InterPro

- ❑ You can use InterPro if you have an amino acid sequence or set of sequences and you want to know:
 - what they are, what family they belong to
 - What domains are present & their function and how it can be explained in structural terms

- ❑ InterPro cannot help you if:
 - you want to perform structural alignment of protein sequences
 - you have a genomic DNA sequence and are interested in gene annotation (intron/exon predictions, identification of promoter regions, etc).

InterPro



InterPro

❑ Home Page: Search (by sequence, by text, domain architecture)

Title: sp|P04637|P53_HUMAN Cellular tumor antigen p53 OS=Homo sapiens OX=9606 GN=TP53 PE=1 SV=4

Job ID: iprscan5-R20240822-175639-0242-57134319-p1m

Length: 393 amino acids

Actions: [Download]

Status: finished

Expires: Thu Aug 29 2024

Protein family membership

p53 tumour suppressor family (IPR002117)

Entry matches to this protein

Representative Domains: P53

Family: p53_tumour_suppressor, CELLULAR TUMOR ANTIGEN P53

Domain: p53_DNA-bd, p53_TAD2, p53_TAD, p53_tetramer...

Overview of domain matches:

- P53 - PF00870
- p53_tumour_suppressor - IPR002117
- p53_DNA-bd - IPR011615
- p53_TAD2 - IPR040926
- p53_transactivation_domain - IPR013872
- p53_tetramerism - IPR010991

Classification of protein families

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Browse / By Entry / InterPro / IPR011615 / Overview

D IPR011615 p53, DNA-binding domain

InterPro entry

Short name: p53_DNA-bd

Overlapping homologous superfamilies:

- p53-like transcription factor, DNA-binding domain superfamily (IPR008967)
- p53/RUNT-type transcription factor, DNA-binding domain superfamily (IPR012346)

Description

P53 is a tumor suppressor gene product; mutations in p53 or lack of expression are found associated with a large fraction of all human cancers. P53 is activated by DNA damage and acts as a regulator of gene expression that ultimately blocks progression through the cell cycle. P53 binds to DNA as a tetrameric transcription factor. In its inactive form, p53 is bound to the ring finger protein Mdm2, which promotes its ubiquitinylation and subsequent proteosomal degradation. Phosphorylation of p53 disrupts the Mdm2-p53 complex, while the stable and active p53 binds to regulatory regions of its target genes, such as the cyclin-kinase inhibitor p21, which complexes and inactivates cdk2 and other cyclin complexes [10, 9, 8, 7, 6, 4, 5, 2, 3, 1].

Overview

Proteins	6k
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CDD

- ❑ [Conserved Domains Database \(CDD\) and Resources \(nih.gov\)](#)
- ❑ CDD is a protein annotation resource that consists of a collection of well-annotated multiple sequence alignment models for ancient domains and full-length proteins.
- ❑ These are available as position-specific score matrices (PSSMs) for fast identification of conserved domains in protein sequences via RPS-BLAST.
- ❑ CDD content includes NCBI-curated domains, which use 3D-structure information to explicitly define domain boundaries and provide insights into sequence/structure/function relationships, as well as domain models imported from a number of external source databases (Pfam, SMART, COG, PRK, TIGRFAMs).